(KNN) K-NEAREST Neighbour

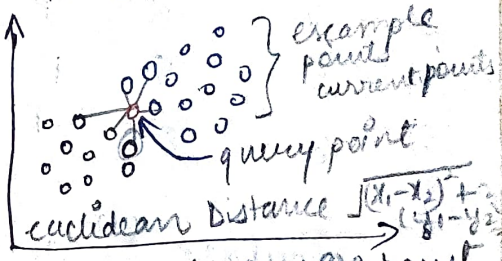
- Identifies data points that are separated into several classes to predict the classification of a new sample data points.

- A Lazy algorithm (memorize the process).
  {does not take its own decision} – no learning.
- Supervised learning approach (easiest algorithm.
- Classifies new points based on a Similarity measures.
  - Euclidean Distance B/W two points.
    - it is a proximity measure and it tries to identify its who ~~alway~~ ~~parents~~ neighbour

other distance – Mantiatan distance or Minkowski distance or Manthatan etc.

How does it works ?

N = 30 K = 3



On the given axis
we have some set of points
suppose we have 30 points
that is 30 vectors or 30 data points.
Some are in Blue, or Red or Black
we choose one point that is we are interested in this point
Now we want to find the neighbourhood

Algorithm

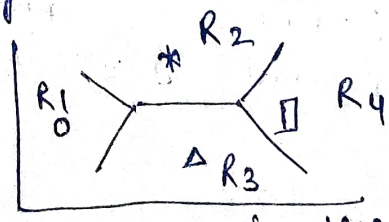
1. Load Data . csv or .xls file.
2. Initialize K → nearest neighbour {K=3, K=5 or K=7}
3. For each sample in the training data
   a) calculate distance B/W query point and current point
   b) Add the distance and the index of the example to an ordered collection.
4. Sort – Ordered collection of distances & indexes from small to large (Ascending Order).
5. Pick first K entries from sorted collection.
6. Get the labels of selected K labels
7. If regression – return mean of K labels
   If classification – Return mode of K label

Remember K parameter

* K must be ODD
* K must not be a multiple of classes.

eg C = 7 x 14 → not considered

if K = 1



So it partitions the space into n no. of Regions called voronoi partition space. If data point belong to R_1 is assigned to class R_1.

... features BMI, Age.

K- NEAREST NEIGHBOUR ALGORITHM (KNN) ②(6)

Example - for Lazy learning — because it will not learn from training data, it memorize the training data

Eg- Exam - Preparation — understand ✓ — memorize →

Example

Given data Query

x (maths = 6 , CS = 8).

and K = 3 — constant tells about nearest neighbour

classification - Pass | fail

	Maths	CS	Result
1)	4	3	F
2)	6	7	P
3)	7	8	P
4)	5	5	F
5)	8	8	P

Euclidean Distance (d)

$$d = \sqrt{(x_{O_1} - x_{A_1})^2 + (x_{O_2} - x_{A_2})^2}$$

O - Observed value eg = O_1 6. O_2 - 8

A - actual value eg = A_1 4. A_2 3

calculate (i) $d_1 = \sqrt{(6-4)^2 + (8-3)^2}$

$= \sqrt{2^2 + 5^2} = \sqrt{29} = 5.38$

(ii) ✓ $d_2 = \sqrt{(6-6)^2 + (8-7)^2} = \sqrt{0^2 + 1^2} = \sqrt{1} = 1$

(iii) ✓ $d_3 = \sqrt{(6-7)^2 + (8-8)^2} = \sqrt{(-1^2) + 0} = 1$

(iv) $d_4 = \sqrt{(6-5)^2 + (8-5)^2} = \sqrt{1^2 + 3^2} = \sqrt{1+9} = \sqrt{10} = 3.16$

(v) ✓ $d_5 = \sqrt{(6-8)^2 + (8-8)^2} = \sqrt{2^2 + 0} = \sqrt{4} = 2$

3 - Pass
0 - fail.

] - Pass is dominating so Problem is classify problem solution as

Pass

Eg 2. Apply K nearest neighbour classifier to predict the diabetic patient with the given features BMI, Age.
if the training examples are -

- Assume K = 3,
- Test example BMI = 43.6, Age = 40, Sugar = ?

$$\text{Distance} = \sqrt{(x_2 - x_1)^2 - (y_2 - y_1)^2}$$

	BMI	Age	Sugar	Rank
1)	33.6	50	1	2
2)	26.6	30	0	
3)	23.4	40	0	
4)	43.1	67	0	
5)	35.3	23	1	
6)	35.9	67	1	
7)	36.7	45	1	1
8)	25.7	46	0	3
9)	23.3	29	0	
10)	31	56	1	

Distance

1) $\sqrt{(43.6-33.6)^2 + (40-50)^2}$
$= 14.14$

2) $= \sqrt{(43.6-26.6)^2 + (30-50)^2} = 19.72$

3) $= \sqrt{(43.6-23.4)^2 + (40-50)^2} = 20.20$

4) $= \sqrt{(43.6-43.1)^2 + (40-67)^2} = 27.00$

5) $= \sqrt{(43.6-35.3)^2 + (40-23)^2} = 18.92$

6) $= \sqrt{(43.6-35.9)^2 + (40-67)^2} = 28.08$

7) $= \sqrt{(43.6-36.7)^2 + (40-45)^2} = 8.52$

8) $= \sqrt{(43.6-25.7)^2 + (40-46)^2} = 18.88$

9) $= \sqrt{(43.6-23.3)^2 + (40-29)^2} = 23.09$

10) $= \sqrt{(43.6-31)^2 + (40-56)^2} = 20.37$

Two = 1
one = 0

Sol^n should be
our case is classified in
class 1 - Sugar.

Sugar = 1.

Eg 3 Solved example for KNN classifier to classify New Instance IRIS Example.

- KNN is a non parametric supervised learning technique in which we try to classify the data point to a given category with the help of training set.

Or. It captures information of all training cases and classify new cases based on a similarity.

" Predictions are made for a new instance (X) by searching through the entire training set for the K most similar cases (neighbour) and summerize the o/p variable for those K cases.
In classification this is the mode (most common) class value."

example